Report_Sam (deadline 30 jun)

"Does Gradient Boosting improve with Fisher's Discriminant analysis? Proposal of an integrated approach and empirical assessment of five popular datasets

Reviewing: 1

Major points:

1) The most significant methodological concern is the tuning strategy. The authors fix GBM hyperparameters and then apply the same parameters to PCA+GBM, LDA+GBM, and LdaBoost — justifying this as a way to isolate the effect of the feature transformation. However, this creates an inherently unequal playing field. The baseline GBM parameters were optimized for the original feature space, which may be suboptimal (or even detrimental) for the transformed spaces. The result is that poor performance of PCA+GBM or LDA+GBM could reflect suboptimal tuning rather than any intrinsic limitation of the method. Only LdaBoost_tuned receives full independent tuning, making cross-method comparisons difficult to interpret cleanly. The authors should either tune all methods independently, or at minimum discuss this limitation more prominently and consistently.

Yes, we agree. We have now fully tuned all parameters, removing the distinction between LdaBoost and LdaBoost_tuned. The updated results are reported in the revised Table 1. As a consequence, a greater heterogeneity across methods emerges (also PCA+GBM outperforming others in one case). However, overall—and excluding the HAR dataset—cross-validation metrics suggest that approaches based on LDA remain promising.

Cross-validated accuracy table

Dataset	GBM	PCA+GBM	LDA+GBM	LdaBoost
HAR	**0.995 ± 0.002**	0.942 ± 0.006	0.978 ± 0.004	0.978 ± 0.006
RAINFALL	0.861 ± 0.025	0.863 ± 0.015	0.860 ± 0.021	**0.864 ± 0.025**
IRIS	0.960 ± 0.044	0.967 ± 0.031	**0.980 ± 0.020**	0.967 ± 0.031
SONAR	0.694 ± 0.173	**0.764 ± 0.016**	0.688 ± 0.052	0.721 ± 0.075
YEAST	0.568 ± 0.037	0.582 ± 0.023	**0.614 ± 0.030**	0.589 ± 0.032

Test validation accuracy table

Holdout setup for this block:

- HAR (official split): train size = 7352, test size = 2947.
- RAINFALL (train split fallback): train size = 1752, test size = 438.

Dataset	GBM	PCA+GBM	LDA+GBM	LdaBoost
HAR	0.9400	0.9053	**0.9528**	0.9511
RAINFALL	**0.8790**	0.8721	0.8607	0.8584

2) While the authors tune the number of PCA components, they retain the maximum number of discriminant components c for LDA (C−1). This is theoretically motivated, but it means PCA is tuned while LDA is not, adding another asymmetry in the comparison. An ablation study retaining fewer LDA components would strengthen the empirical analysis.

SI PUO FARE rispetto a TABLE 1 COSA VIENE AL VARIARE per GBM+LDA (al variare di numero di var discriminanti PER Yeast (10 classi c=1,5, 9 già c'è nel paper) e HAR (6 classi, c=1, 3, 5 già c'è nel paper)?

3) The paper relies on cross-validated means and standard deviations to compare methods, but never performs any formal statistical tests (e.g., Wilcoxon signed-rank test, Friedman test, or corrected paired t-tests). Given the overlapping standard deviations visible in Table 1, many of the claimed improvements may not be statistically significant. This is a standard expectation in comparative machine learning studies.

We add a note indicating that results based on the test set (Wilcoxon signed-rank test) show no statistically significant differences among the accuracies.

4) LdaBoost refits LDA at every boosting iteration, which is considerably more expensive than standard GBM. The paper contains no discussion whatsoever of computational overhead — training times, scalability to large datasets, or memory footprint. This is a major omission for a method intended for practical use.

At the end of the result section, we provide empirical evidence on the computational time of LdaBoost in comparison with standard GBM (added Table A1 in the Appendix).

Testo tuo da scrivere prima del paragrafo **Comparative performance of different pipelines: simulations**

Table A1 in the Appendix reports the runtime of the considered methods across the different datasets. Qui rivedi ed organizza i concetti sotto

Lda+GBM e l'LdaBoost su dataset molto grandi come l'HAR ci mettono nettamente meno tempo (LDA+GBM ha uno speedup di 65,8 rispetto alla GBM)

From a computational perspective, the LDA-based approaches remain very competitive while keeping strong accuracy. One-shot LDA+GBM is generally the fastest option (and in HAR it is dramatically faster than PCA+GBM), while still delivering high predictive performance. At the same time, LdaBoost deserves a positive

Da aggiungere in Appendix

Table A1: Mean runtime by dataset and method ….una sola tabella non due

Mean runtime by dataset and method

Dataset	PCA+GBM	LDA+GBM	LdaBoost	GBM
HAR	00:06:47	00:00:09	00:03:24	00:07:09
RAINFALL	00:00:01	00:00:01	00:00:02	00:00:00
IRIS	00:00:00	00:00:00	00:00:02	00:00:00
SONAR	00:00:00	00:00:00	00:00:02	00:00:00
YEAST	00:00:03	00:00:03	00:00:16	00:00:02

Test validation timing table

Method	HAR Runtime (s)	RAINFALL Runtime (s)
GBM	675.07	0.53
LDA+GBM	10.25	0.27
PCA+GBM	652.62	0.70
LdaBoost	286.22	2.33

5) In binary classification (C=2), LDA produces exactly one discriminant component. Training GBM trees on a single feature per iteration is a severe constraint, and it is somewhat surprising LdaBoost performs competitively at all. The authors should discuss this case more carefully, since the theoretical motivation for LdaBoost is much more compelling in the multiclass setting.

We thank the reviewer for this insightful comment. We agree that in the binary classification setting, Fisher Discriminant Analysis yields a single discriminant component, which may appear to strongly constrain the subsequent GBM model, since all trees are effectively built on a one-dimensional representation. However, we would like to clarify why this does not necessarily limit the effectiveness of the proposed approach.

First, the approach is *supervised* and explicitly optimized to maximize class separation, also in the binary case, where the resulting feature can be interpreted as a highly informative summary of the original feature space, rather than a generic projection.

Second, even when operating on a single feature, GBM with decision trees retains a significant degree of flexibility. Trees can model highly nonlinear decision boundaries along this one-dimensional projection by introducing multiple splits, effectively approximating complex threshold-based decision rules. In other words, while the input space is reduced, the functional space explored by boosting remains rich due to the nonlinearity of the base learners.

Third, the binary case can be interpreted as a setting where the method trades representational dimensionality for noise reduction and stability. By collapsing the input space onto the most discriminative direction, the model may benefit from reduced variance and improved generalization, especially in high-dimensional or noisy settings. This may explain why the method performs competitively despite the apparent restriction.

We agree, however, that the theoretical motivation is more compelling in the multiclass setting, where FDA produces multiple discriminant components and the boosting stage can exploit a richer representation. In the discussion (new section) we mention this issue for binary case, clarifying both its limitations and its potential advantages, and to better position the method within this context.

Testo aggiunto nella discussion (nuova sezione 5)

6) The related work focuses on feature extraction literature but does not engage with the rich literature on boosting variants. Methods such as LPBoost, FilterBoost, or boosting with kernel projections bear at least some conceptual similarity to LdaBoost. A brief discussion of how LdaBoost relates to or differs from these would strengthen the paper's positioning.

In the introduction, we present these methods with particular emphasis on their use of features, specifically whether they rely on original features or on features derived through supervised or unsupervised approaches.

Testo (da mettere nella intro, prima della frase "While not standard, the use of PCA")

Linear Programming Boosting -LPBoost; \citep{demiriz2002lpboost} and FilterBoost \citep{bradley2008filterboost} are two variants of classical boosting that primarily modify how weak learners are selected and weighted. LPBoost reformulates boosting as a global optimization problem, closely related to margin-based methods such as support vector machines, with the goal of finding an optimal and often sparse combination of weak learners rather than adding them in a purely greedy, sequential manner as in Gradient Boosting Machines (GBM). FilterBoost, by contrast, is designed with scalability in mind: it introduces a filtering or sampling mechanism that allows the algorithm to operate on subsets of the training data at each iteration, thereby reducing computational cost while maintaining predictive performance. In both cases, however, the representation of the data remains unchanged. The algorithms operate directly on the original input features, without introducing new feature transformations beyond standard preprocessing steps. The key differences with respect to GBM therefore lie in the learning strategy rather than in the feature space.

Boosting with Kernel Projections \citep{kegl2009kernelboost} introduces a more substantial departure from the classical framework by incorporating a transformation of the input space prior to or within the boosting procedure. The central idea is to map the original features into a higher-dimensional or more expressive space through a kernel-based transformation, enabling the construction of weak learners that can capture more complex, nonlinear patterns. The boosting process then proceeds in this transformed space, iteratively combining weak learners to improve predictive accuracy on the target. Conceptually, this approach combines the representational power of kernel methods with the sequential error-correcting nature of boosting.

From the perspective of feature representation, this third approach differs fundamentally from both GBM and its variants such as LPBoost and FilterBoost. The transformed features are not simple linear combinations of the original inputs but arise from a kernel-induced mapping that may be implicit and high-dimensional. Importantly, this mapping is typically not constructed in a supervised manner with respect to the target; it is defined a priori by the choice of kernel function. However, the boosting procedure introduces a form of indirect supervision by progressively selecting and emphasizing those components of the transformed space that are most useful for predicting the target. In this sense, feature construction occurs in two stages: an initial, largely unsupervised expansion of the feature space, followed by a supervised selection driven by the

boosting iterations. More advanced variants further blur this distinction by adapting or combining kernels during training, thereby allowing the representation itself to become partially target-dependent. This dual mechanism distinguishes kernel-based boosting approaches from classical GBM, where both representation and learning are typically confined to the original feature space.

7) The simulation study is a welcome addition, but the data-generating process (multivariate normal with class-specific means) is actually the setting where LDA is asymptotically optimal by design. Results here are therefore somewhat expected and may not generalize to real data with non-Gaussian features, nonlinear class boundaries, or mixed variable types. The authors should acknowledge this limitation and ideally complement the simulations with non-Gaussian scenarios.

We thank the reviewer for these important observations. We agree that the simulation setting based on multivariate normal distributions with class-specific means corresponds to a scenario in which LDA is asymptotically optimal, and therefore provides a favorable context for methods that build upon LDA. Our primary goal in this initial study was to introduce LdaBoost and evaluate its behavior in a controlled setting where the underlying structure is well understood, allowing for a clear comparison with the Bayes optimal classifier. In this sense, the chosen data-generating process serves as a benchmark to assess whether the proposed method can recover near-optimal performance under ideal conditions.

We acknowledge, however, that such results may not directly generalize to more complex and realistic data settings, including non-Gaussian distributions, nonlinear class boundaries, or mixed-type features. We will revise the manuscript to explicitly state this limitation in the new discussion section (5. Discussion). Extending the simulation study to a broader range of data-generating mechanisms is an important direction for future work, and will be considered in subsequent studies to further assess the robustness and practical applicability of LdaBoost.

Testo aggiunto nella discussion

8) The fact that baseline GBM outperforms LdaBoost on cross-validated accuracy for the largest and most feature-rich dataset (HAR, 561 features, 10,299 observations) is actually the most informative result in the paper, yet it receives relatively little discussion. This is the setting where one might most expect LDA preprocessing to help with dimensionality reduction — why does it not? A more thorough analysis of this "failure mode" would be valuable.

We include in the second part of the Discussion some hypotheses to explain this behavior.

Testo aggiunto nella discussion

Minor points

- Figure captions are inconsistent: some figures in the body and appendix share nearly identical descriptions, making it hard to distinguish them.
- The abstract claims results "show that applying LDA... significantly improves classification accuracy" — the word "significantly" implies statistical testing, which is not performed.

CORREGGI E LEVA OVERALL significantly O significant

- The paper would benefit from a short limitations section before or in the conclusion.

We add limitations in the last part of the discussion.

Reviewing: 2

Major Concerns

1. Page 7 of 21, lines 22 through 27: The text implies that LDA is recalculated at each boosting round. However, the Fisher criterion is based on maximizing between-class scatter relative to within-class scatter. If residuals replace labels, the theoretical justification for applying LDA in this way is not obvious. The manuscript should explain how the optimization problem remains valid under residual-based updates.

3. Steps 2 and 3 in the step-by-step algorithmic implementation of the LdaBoost procedure look fine and follow standard boosting ideas. The main concern is Step 1. LDA normally works with fixed class labels, but here the method tries to relabel samples using residuals and then refit LDA (as mentioned in 1). Residuals are not the same as labels, so it is unclear if this is a valid use of LDA. The validity of this step is unclear without additional justification. Hence, without a clear explanation, this part of the method feels uncertain. The authors should explain more clearly how residuals can be used in LDA and why this makes sense.
4. Step 4 in the algorithmic implementation of the LdaBoost procedure itself looks fine. However, it relies on Step 1 because the scores it updates come from that LDA step. If Step 1 is not clearly explained or justified, then Step 4 might end up relying on something that is not solid or reliable.

We thank the reviewer for this thoughtful comment and for highlighting an important aspect of the proposed method. We agree that classical LDA is defined for fixed class labels, and that residuals are not labels in the usual sense. However, we would like to clarify that in our approach LDA is not applied directly to continuous residuals.

Our procedure follows the general logic of gradient boosting. In classical GBM for categorical targets, subsequent learners are not trained on the original class labels, but on pseudo-residuals, i.e., the negative gradients of the loss function with respect to the current model (Friedman, 2001). These quantities are continuous and differ substantially from the original labels, yet they provide the supervision signal that guides the learning process.

In our method, we use these gradient signals to construct *pseudo-labels*: after computing the residuals, each observation is reassigned to the class corresponding to the largest positive gradient component. This step can be interpreted as a discretization of the functional gradient, identifying for each observation the direction in which the model most needs to adjust its prediction. LDA is then applied to these dynamically relabeled data, rather than to the residuals themselves.

From this perspective, the procedure remains consistent with the functional gradient descent view of boosting (Friedman, 2001). Moreover, the relabeling step can be related to cost-sensitive and reweighted classification frameworks, where the learning problem is adapted to emphasize difficult observations (Elkan et al., 2001, Zadrozny at al., 2003), as well as to hard assignment strategies derived from soft signals, as in expectation–maximization (EM) algorithms (Dempster et al., 1977).

Overall, the role of LDA in this framework is to identify, at each iteration, a discriminant subspace that emphasizes the directions where the model is currently making the largest errors, as indicated by the gradient. We will add this fundamental point in the first part of the discussion section to clarify this point and make the distinction between residuals and pseudo-labels more explicit.

Testo aggiunto nella prima parte della discussion

2. In the LdaBoost description, the authors recomputed LDA at each boosting round, a procedure that can become computationally expensive, particularly in high-dimensional settings. Please add few sentences discussing the scalability of this iterative approach and compare it with using LDA once as a pre-processing step.

At the end of the result section, we provide empirical evidence on the computational time of LdaBoost in comparison with standard GBM (added Table A1 in the Appendix).

Minor Concerns

• Pages 6 of 21 lines 28:29: The statement "predictors with larger dissimilarities between class means will have larger weights" is oversimplified. In LDA, feature weights depend on both how far apart class means are and how consistent the feature is within each class. Please revise this explanation to reflect both components of the Fisher criterion so that the description is accurate and balanced.
CORREGGI

• Minor typographical errors (e.g., "repored" on page. 9 of 21 line 3).
CORREGGI

• Page 12 of 21 line 14 through 19: Please consider revising the sentence. You can start the sentence with "We first begin our simulations. . . "Also, please note that sample of 1,000 is not typically considered "small," especially in simulation studies.

CORREGGI

This manuscript has potential to contribute to the literature on research in integrated learning models that integrate classical statistical discriminant methods with modern ensemble algorithms to improve classification performance. Fixing the methods section and a few smaller issues as mentioned above will significantly improve its rigor and readability. I recommend revision at this stage.

TESTO DELLA DISCUSSION COMPLETA + bibtex

5. Discussion

This section discusses the rationale and interpretation of the proposed LdaBoost method, with particular attention to its learning mechanism and its relationship to the gradient boosting framework.

Firstly, the proposed approach builds upon the general principles of gradient boosting. In classical GBM for categorical targets, subsequent learners are not trained directly on the original class labels, but on pseudo-residuals, i.e., the negative gradients of the loss function with respect to the current model {friedman2001gbm}. These quantities are continuous and differ substantially from the original labels, yet they provide the supervision signal that drives the iterative refinement of the model.

In our method, rather than applying LDA directly to these continuous residuals, we use the gradient information to construct *pseudo-labels*. Specifically, after computing the residuals, each observation is reassigned to the class corresponding to the largest positive gradient component.

This step can be interpreted as a discretization of the functional gradient, identifying, for each observation, the direction in which the model most needs to adjust its prediction. LDA is then applied to these dynamically relabeled data, thereby preserving its natural formulation as a method for supervised dimensionality reduction with categorical labels. In other words, the pseudo-label identifies the class toward which the model "should move" in order to correct its current error.

This procedure introduces a form of dynamic supervision, in which the labels are no longer fixed but evolve across iterations, reflecting the current state of the model. As a consequence, LDA is not used to separate the original classes, but rather to identify, at each iteration, the directions in the feature space that best discriminate among the required corrections. In this sense, the resulting discriminant projection is locally adapted to the model's errors and allows the learning process to focus on the most challenging regions of the problem.

From this perspective, the relabeling step can be viewed in connection with cost-sensitive and reweighted classification frameworks, where the learning problem is modified to emphasize difficult or misclassified observations \citep{elkan2001cost, zadrozny2003cost}, as well as with hard assignment strategies derived from soft signals, as in expectation–maximization (EM) algorithms \citep{dempster1977em }. In this end, LDA plays a dynamic role within this framework: at each iteration, it identifies a discriminant subspace that emphasizes the directions where the model is currently making the largest errors, as indicated by the gradient signal.

Second, in the binary classification setting, Fisher Discriminant Analysis yields a single discriminant component, and it may therefore appear somewhat surprising that LdaBoost performs competitively at all given the resulting one-dimensional representation. However, this component is obtained through a supervised procedure explicitly designed to maximize class separation, and therefore represents a highly informative projection of the original feature space. Rather than constituting a generic dimensionality reduction, it can be interpreted as a targeted feature extraction step that concentrates the relevant signal for the prediction task into a single direction.

Even when operating on a one-dimensional representation, Gradient Boosting Machines with decision trees retain considerable modeling flexibility. Decision trees can introduce multiple splits along the projected axis, enabling the approximation of complex and nonlinear relationships through combinations of threshold-based rules. As a result, the restriction to a single feature does not necessarily imply a substantial loss in the expressive power of the overall model.

From a statistical perspective, the binary case can also be viewed as a setting in which the method trades representational richness for noise reduction and stability. By projecting the data onto the most discriminative direction, the approach may reduce variance and improve generalization, particularly in high-dimensional or noisy scenarios. This perspective helps explain why competitive empirical performance can still be observed despite the apparent constraint. At the same time, the advantages of the proposed approach become more pronounced in the multiclass setting, where multiple discriminant components provide a richer representation for the subsequent boosting stage.

Third, despite the strong predictive accuracy of LDA-based methods, an important finding is that GBM (using cross-validated metrics) outperforms the other approaches on the largest and most feature-rich HAR dataset. A possible explanation is that, in this setting, the supervised dimensionality reduction induced by LDA may compress the original information too aggressively. In particular, LDA produces at most $C-1$ discriminant components, thereby representing the data through a small number of global linear directions; this can be limiting when the discriminative signal is distributed across many variables or depends on nonlinear relationships among features \citep{hastie2009elements}. By contrast, GBM can naturally capture threshold effects, feature interactions, and complex nonlinear structures through ensembles of decision trees, making it particularly effective in high-dimensional settings with heterogeneous features \citep{friedman2001gbm}. This suggests that, for rich datasets such as HAR, preserving the original feature space may offer a substantial advantage.

Moreover, the literature highlights several well-known limitations of LDA/FDA: performance may degrade when the Gaussian class-conditional assumption and the homogeneity of covariance matrices are violated \citep{hastie2009elements}; when classes exhibit multimodal structure that cannot be captured by a single discriminant direction \citep{sugiyama2007lda}; and in high-dimensional regimes, where noise accumulation can adversely affect linear discriminants \citep{fan2008highdim}. These factors provide a plausible explanation for the observed behavior on the HAR dataset. Notice, however, that LdaBoost outperforms GBM on the test set for the HAR dataset.

The paper has also limitations.
Firstly, the simulation study is based on a multivariate normal data-generating process with class-specific means, a setting in which Linear Discriminant Analysis is known to be asymptotically optimal. As a consequence, the proposed LdaBoost method is evaluated in a scenario that is particularly favorable to its underlying structure, and the observed performance may therefore be partly expected. While this choice allows for a clear and interpretable comparison with the Bayes optimal classifier in a controlled framework, it does not fully reflect the complexity of real-world data, which often exhibit non-Gaussian distributions, nonlinear class boundaries, and heterogeneous feature types. Future work will extend the empirical analysis to more general settings, including non-Gaussian and nonlinear scenarios, in order to better assess the robustness and generalizability of the proposed approach.
Second, since LDA is traditionally defined for fixed class labels, the use of LdaBoost—where samples are relabeled based on residuals and LDA is iteratively refitted—may appear as a heuristic attempt to extend LDA to a setting that is more naturally suited to continuous targets, as in standard GBM. In the Discussion section, we provide a theoretical justification for this procedure, grounded in the gradient boosting framework of loss function minimization, as well as in related machine learning approaches that rely on pseudo-labels derived from residuals.

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